

Cybersecurity Task-1 Threat Report (Awareness & Research Project)

Internship Program Name	Cybersecurity and Ethical Hacking Internship
Task Number	Task-1
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1. Introduction to Cybersecurity

- Cybersecurity is the practice of protecting digital devices, networks and sensitive data from cyber threats such as hacking, malware & phishing attacks.
- It involves using specialized tools to detect and remove harmful software, also a range of strategies, technologies and best practices designed to safeguard computer, networks, data from cyber attacks.
- Importance of Cybersecurity for Individuals and Businesses: -
 - Protects Personal Information:
 - Cybersecurity helps protect personal data such as passwords, Bank details, Aadhaar/PAN information, Social media accounts, Photos and private files.
 - Without proper security, attackers can steal or misuse this information.
 - Maintains Customer Trust:
 - Customers trust companies that keep their data secure. A data breach can damage a company's reputation and reduce customer confidence.
 - Protects Privacy:
 - Individuals use the internet daily for communication, shopping, and social networking.
 - Cybersecurity prevents unauthorized access to personal accounts and private conversations.
 - Prevents Financial Damage:
 - Cyberattacks can cause loss of money, System downtime, Recovery costs, Legal penalties.
 - Strong cybersecurity reduces these risks.
- Current relevance of cybersecurity: -
 - Cybersecurity has become extremely important in the modern digital world due to rapid increase in cyber-crimes, growing dependence on technology, and the emergence of AI-driven threats.
 - Individuals, businesses, governments, and organizations rely heavily on digital systems for communication, banking, healthcare, education, and businesses operations.

1) Increasing Cyber Crimes:

- Cybercrime cases re increasing worldwide every year. Attackers use advanced methods to steal data, hack systems, spread malware, and perform online fraud. Common cybercrimes include:
 - ✓ Phishing attacks
 - ✓ Identity theft
 - ✓ Ransomware attacks
 - ✓ Banking fraud
 - ✓ Social media hacking
 - ✓ Data breaches
- Cybercriminals target both individuals and organizations to gain financial benefits or sensitive information.
- Example- Ransomware attacks on hospitals and companies have caused major financial losses and service disruptions in recent years.

2) Growing Digital Dependency:

- Modern society depends heavily on digital technology and internet-based services. Important activities such as:

- ✓ Online banking
 - ✓ E-commerce
 - ✓ Cloud storage
 - ✓ Remote work
 - ✓ Digital payments
 - ✓ Social networking
 - ✓ Online education
 - All are performed through connected systems and networks.
 - Because of this dependency, any cyberattack can affect daily life, business operations, and even national security
- 3) AI-driven Threats:
- Artificial Intelligence (AI) is improving technology, but cybercriminals are also using AI to create more advanced attacks.
 - Examples of AI-Based Threats:
 - AI-generated phishing emails
 - Deepfake videos and voice scams
 - Automated hacking tools
 - Intelligent malware
 - Password-cracking systems
 - Example- Deepfake technology can imitate a person's voice or face to perform scams or spread misinformation.

2. 5 Major Modern Cyber Threats

a. AI-Powered Phishing Attacks: -

- ❖ AI-powered phishing attacks use Artificial Intelligence to create highly convincing fake emails, websites, voice messages, and deepfake videos.
- ❖ These attacks are more dangerous because AI removes common mistakes such as poor grammar, and unrealistic formatting, making scams difficult to identify.
- ❖ Attackers use:
 - ✓ AI-generated emails
 - ✓ Deepfake voice/video scams
 - ✓ Social engineering
 - ✓ Fake login pages
 - ✓ Automated phishing kits
- ❖ Latest example-
In 2026, the FBI warned about a phishing toolkit called Kali365, which used AI-generated phishing emails and fake Microsoft login pages to steal Microsoft 365 authentication tokens. Attackers could bypass Multi-Factor Authentication (MFA) and access Outlook, OneDrive, and Teams accounts.
- ❖ Prevention:
 - Use Multi-Factor Authentication (MFA)
 - Verify URL's carefully
 - Avoid checking suspicious links
 - Conduct employee awareness training
 - Use AI-based email security filters



Figure 1. AI-Powered Phishing Attack Workflow

b. Ransomware-as-a-Service (RaaS): -

- ❖ Ransomware-as-a-Service is a cybercrime business model where ransomware developers sell or rent ransomware tools to criminals.
- ❖ Even low-skilled attackers can launch ransomware attacks using ready-made kits.
- ❖ These attacks:
 - ✓ Encrypt files
 - ✓ Demand ransom payments
 - ✓ Disrupt operations
 - ✓ Cause financial losses
- ❖ Latest example-
In 2025-2026, group such as RansomHub, BlackByte, Akira, and Qilin became highly active in the RaaS ecosystem. Cybercriminal services were professionally organized and targeted companies worldwide. Microsoft also disrupted the Fox Tempest cybercrime network that distributed ransomware using fake software installers and signed malware certificates.
- ❖ Prevention:
 - Regularly backup important data
 - Update systems and software
 - Use endpoint protection tools
 - Restrict unauthorized access
 - Train employees against phishing attacks

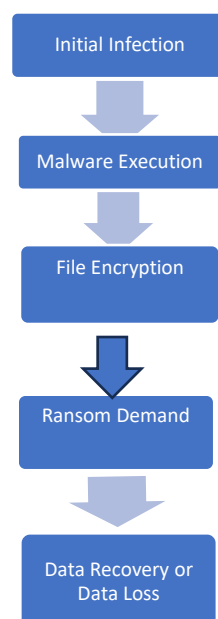


Figure 2. Ransomware Attack Lifecycle

c. Cloud Security Misconfigurations: -

- ❖ Cloud misconfiguration occurs when cloud services such as AWS, Google Cloud (GCP), or Microsoft Azure are incorrectly configured. This can expose sensitive databases, storage buckets, API keys, or internal systems to the public internet.
- ❖ Common issues include:
 - Open storage buckets
 - Weak access permissions
 - Unencrypted databases
 - Publicly exposed cloud resources
- ❖ Latest example-
A 2025 cloud security report by tenable.com found that:
 - 9% of publicly accessible cloud storage contained sensitive data
 - 97% of exposed cloud data was confidential or restricted

Reports also showed:

- AWS misconfigured resources: 45%
 - GCP misconfigured resources: 63%
 - Azure misconfigured resources: 70%
- ❖ Prevention:
 - Regular cloud security audits
 - Use least-privilege access policies
 - Enable encryption
 - Monitor cloud resources continuously

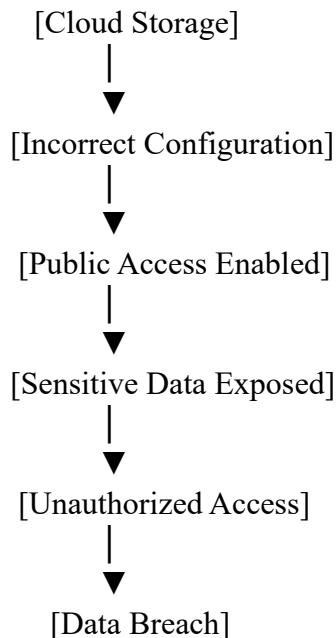


Figure 3. Cloud Security Misconfiguration Attack Flow

d. IoT Vulnerabilities: -

- ❖ Internet of Things (IoT) devices include:
 - Smart cameras
 - Smart TVs
 - Smart home devices

- Industrial sensors
- ❖ Many IoT devices have weak security, default passwords, or outdated firmware, making them easy targets for hackers.
- ❖ Attackers can spy on users, steal data, control devices remotely, use devices in botnet attacks.
- ❖ Latest example-
Modern botnets continue targeting vulnerable smart devices to launch DDoS attacks and network intrusions. Security researchers have warned that poorly secured IoT ecosystems remain a major threat because millions of connected devices are deployed without proper updates or protections.
- ❖ Prevention:
 - Change default passwords
 - Update device firmware regularly
 - Disable unused features
 - Use secure Wi-Fi networks
 - Separate IoT devices from main networks

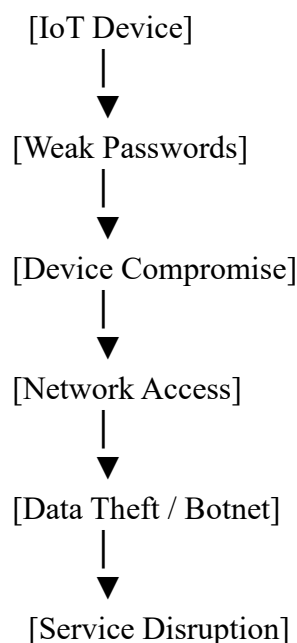


Figure 4. IoT Vulnerabilities

e. Zero-Day Exploits: -

- ❖ A Zero-Day exploits targets a software vulnerability that is unknown to the software vendor or has no security patch available yet.
- ❖ Attackers exploit these vulnerabilities before developers can fix them.
- ❖ Zero-Day attacks are dangerous because:
 - No immediate protection exists
 - Traditional security tools may fail
 - Attackers gain unauthorized access quickly
- ❖ Latest example-
In 2026, researchers reported one of the first known AI-assisted attacks targeting Two-Factor Authentication systems using AI-generated zero-day techniques embedded in malware scripts.

Cybersecurity experts also observed AI-powered malware capable of dynamically modifying its behaviour during attacks to evade detection systems.

❖ Prevention:

- Keep systems updated
- Apply security patches immediately
- Use intrusion detection systems
- Monitor unusual activity
- Use advanced endpoint security solutions

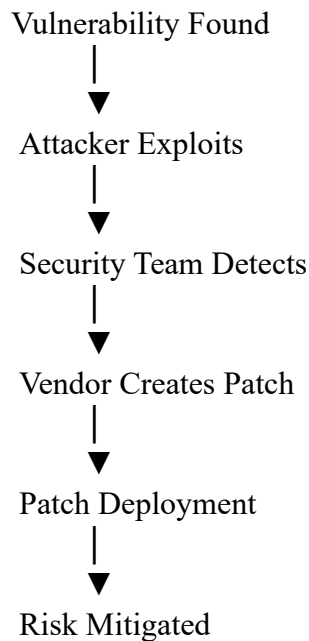


Figure 5. Timeline showing window of exposure during zero-day exploit

3. Impact Analysis

For each threat, it can harm Individuals (data theft, financial fraud) and Organizations (loss of data, downtime, reputation damage, compliance issues).

1. AI-Powered Phishing Attacks-

i. Impact on Individuals:

- Theft of personal information such as passwords, banking details, and OTPs.
- Financial fraud through fake banking or payment links.
- Identity theft and unauthorized access to social media or email accounts.
- Emotional stress and privacy loss due to account compromise.

Common Causes of Data Breaches

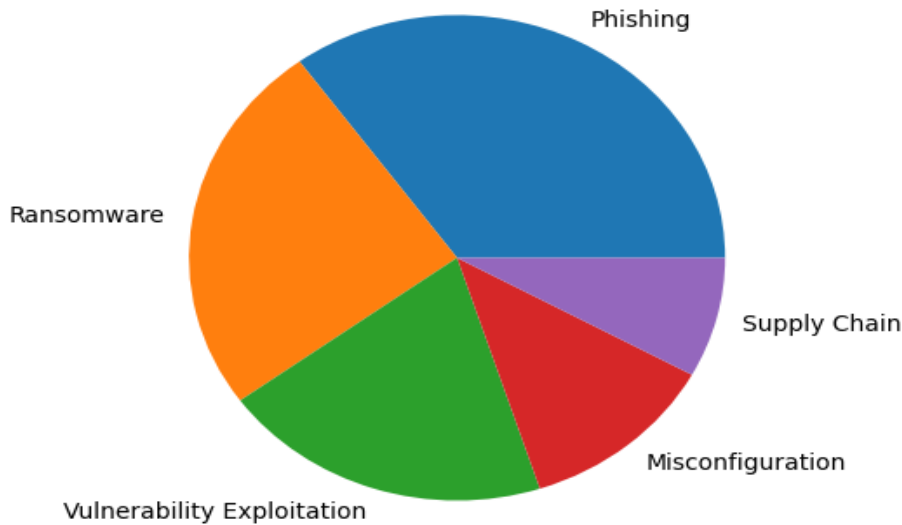


Figure 6. Common Causes of Data Breaches

ii. Impact on Organizations:

- Data breaches caused by stolen employee credentials.
- Unauthorized access to company systems and cloud accounts.
- Financial losses from fraud or ransomware deployment after phishing attacks.
- Damage to organizational reputation and customer trust.

2. Ransomware-as-a-Service (RaaS)-

A. Impact on Individuals:

- Loss of personal files, photos, and important documents.
- Financial loss due to ransom payments.
- Exposure of sensitive personal data if attackers leak information.
- Disruption of daily digital activities.

B. Impact on Organizations:

- Complete shutdown of business operations and services.
- Loss of important organizational data through encryption or deletion.
- High recovery costs and ransom payments.
- Downtime affecting productivity and customer services.
- Legal penalties for failing to protect sensitive information.

3. Cloud Security Misconfigurations-

i. Impact on Individuals:

- Leakage of personal information stored in cloud platforms.
- Exposure of passwords, documents, or private images.
- Increased risk of identity theft and online fraud.

ii. Impact on Organizations:

- Exposure of confidential company and customer data.

- Unauthorized public access to databases or cloud storage.
- Financial losses from data breaches and recovery processes.
- Compliance violations related to data protection laws.

4. IoT Vulnerabilities-

A. Impact on Individuals:

- Unauthorized access to smart home devices such as cameras and speakers.
- Privacy invasion through spying or monitoring.
- Theft of personal information from connected devices.
- Increased risk of cyberattacks through insecure devices.

B. Impact on Organizations:

- Compromise of connected industrial or office devices.
- Disruption of operational technology systems.
- Use of vulnerable IoT devices in botnet or DDoS attacks.
- Security risks to internal networks and sensitive data.

5. Zero-Day Exploits-

i. Impact on Individuals:

- Devices may be hacked before security updates are available.
- Theft of personal data and sensitive information.
- Malware infections causing system damage or data loss.

ii. Impact on Organizations:

- Attackers can gain unauthorized access to systems and networks.
- Data breaches and loss of confidential business information.
- High costs for incident response and system recovery.
- Serious reputation and compliance issues because attacks are difficult to detect initially.

4. Real-World Case Studies

Using the Verizon Data Breach Investigations Report (DBIR) as a reference, following are the real-world case studies for modern cyber threats: -

(1) AI-Powered Phishing Attacks-

CASE STUDY: MGM Resorts Cyberattack (2023)

- ✓ Attackers used social engineering and AI-assisted techniques to impersonate employees and trick IT support into granting access.
- ✓ Systems across MGM hotels and casinos were disrupted for several days.
- ✓ Customer data was exposed, and the company suffered significant financial losses.

Impact on individuals and organizations: -

- Personal information theft and identity fraud.
- Service disruption, revenue loss, and reputational damage

(2) Ransomware-as-a-Service (RaaS)-

CASE STUDY: WannaCry Ransomware Attack (2017)

- ✓ WannaCry infected more than 200,000 computers across 150 countries.
- ✓ It encrypted files and demanded Bitcoin payments for decryption.
- ✓ Major victims included the UK's National Health Service (NHS), causing widespread operational disruptions.

Impact on individuals and organizations: -

- Loss of access to personal files and financial losses.
- Operational downtime, recovery costs, and business interruption.

(3) Supply Chain Attack-

CASE STUDY: SolarWinds Supply Chain Attack (2020)

- ✓ Attackers compromised SolarWinds Orion software updates.
- ✓ Malicious code was distributed to approximately 18,000 customers.
- ✓ Victims included government agencies and major corporations.

Impact on individuals and organizations: -

- Potential exposure of personal data.
- Long-term espionage, data theft, and significant remediation costs.

(4) Cloud Misconfiguration-

CASE STUDY: Capital One Data Breach (2019)

- ✓ A misconfigured cloud firewall allowed an attacker to access sensitive customer information.
- ✓ Over 100 million customer records were exposed.

Impact on individuals and organizations: -

- Exposure of personal and financial information.
- Regulatory penalties, lawsuits, and reputational damage.

(5) Vulnerability Exploitation (Zero-Day/Unpatched Systems)-

CASE STUDY: Log4Shell Vulnerability (2021)

- ✓ A critical vulnerability in the Log4j logging library affected millions of systems worldwide.
- ✓ Attackers exploited unpatched systems to gain unauthorized access and deploy malware.

Impact on individuals and organizations: -

- Data theft and privacy risks.
- System compromise, ransomware infections, and costly patching efforts

5. Preventive Measures

I. AI-Powered Phishing Attacks-

Preventive Measures:

- Multi-Factor Authentication (MFA): Adds an extra layer of security even if credentials are stolen.
- Security Awareness Training: Educates users to identify phishing emails, fake websites, and deepfake scams.
- Email Security Solutions: Deploy anti-phishing filters, spam detection, and email authentication protocols such as SPF, DKIM, and DMARC.

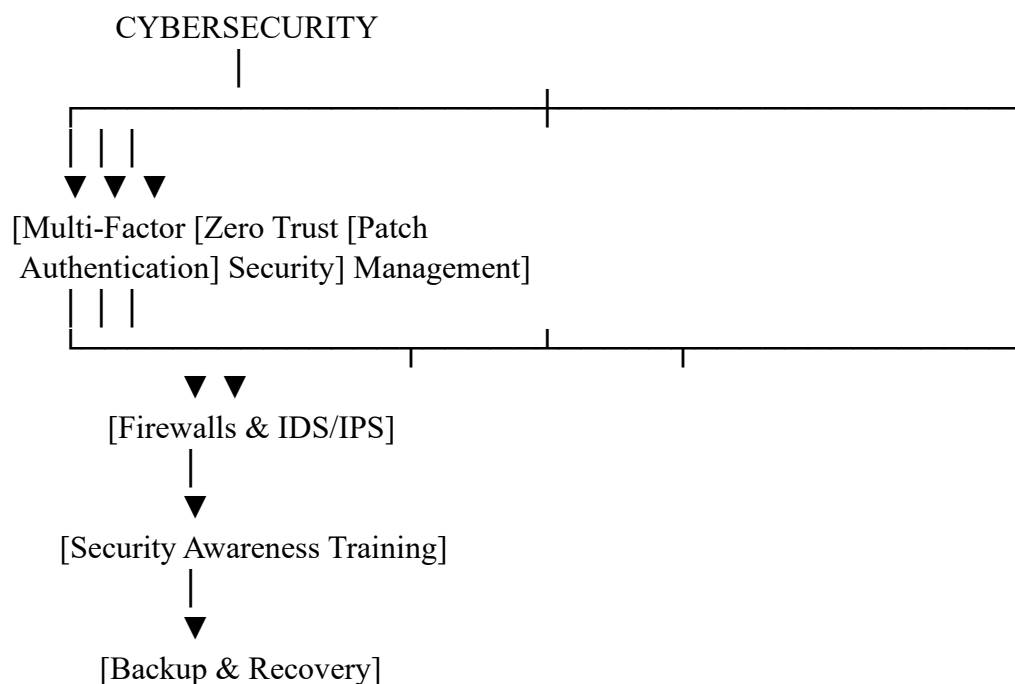


Figure 7. Cyber Security Defense Framework

II. Ransomware-as-a-Service (RaaS)-

Preventive Measures:

- Regular Data Backups: Maintain offline and cloud backups to restore data without paying ransom.
- Endpoint Detection and Response (EDR): Detects and blocks ransomware activity in real time.
- Patch Management: Regularly update operating systems and applications to eliminate exploitable vulnerabilities.

III. Supply Chain Attacks-

Preventive Measures:

- Vendor Risk Assessment: Evaluate the security practices of third-party suppliers and partners.
- Zero Trust Security Model: Verify every user, device, and application before granting access.
- Software Integrity Verification: Use code signing, digital signatures, and Software Bill of Materials (SBOM) to ensure software authenticity.

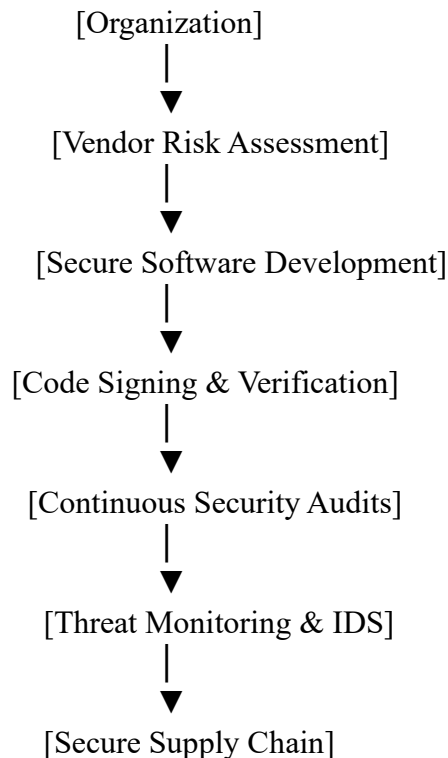


Figure 8. Supply Chain Attack Prevention Framework

IV. Cloud Misconfigurations-

Preventive Measures:

- Cloud Security Audits: Regularly review cloud settings and access permissions.
- Identify and Access Management (IAM): Implement least-privilege access controls.
- Cloud Security Monitoring Tools: Use automated tools to detect misconfigurations and suspicious activities.

V. Vulnerability Exploitation (Zero-Day/Unpatched Systems)-

Preventive Measures:

- Regular Patch Management: Apply security updates and patches promptly.
- Intrusion Detection/Prevention Systems (IDS/IPS): Monitor and block malicious activities.
- Vulnerability Assessments and Penetration Testing: Regularly identify and remediate security weaknesses.

Security Awareness
Access Control & MFA
Firewalls & IDS/IPS
Endpoint Protection
Data Encryption
Backup & Recovery

Figure 9. Layers of Cybersecurity Protection

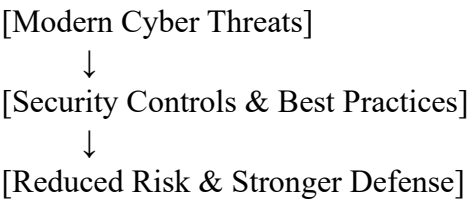
6. Conclusion & Future Scope

a) CONCLUSION

- Cybersecurity has become a critical requirement in today’s digital world, where individuals and organizations rely heavily on technology and online services.
- Modern threats such as AI-powered phishing, ransomware, supply chain attacks, cloud misconfigurations, and vulnerability exploitation can cause significant financial losses, data breaches, operational disruptions, and reputational damage.
- Therefore. Adopting proactive cybersecurity measures such as Multi-Factor Authentication (MFA), regular patch management, employee awareness training, and Zero trust security is essential to reduce risks and protect valuable assets.

b) FUTURE SCOPE

- As technology continues to evolve, cyber threats are becoming more sophisticated and difficult to detect.
- Emerging technologies such as Artificial Intelligence (AI), cloud computing, the Internet of Things (IoT), and quantum computing will introduce new security challenges.
- Therefore, continuous learning, regular security assessments, and staying updated with the latest threat intelligence are crucial for cybersecurity professionals and organizations.
- Investing in advanced security solutions, automation, and ongoing cybersecurity education will help build stronger defences against true cyber threats and ensure a safer digital environment.



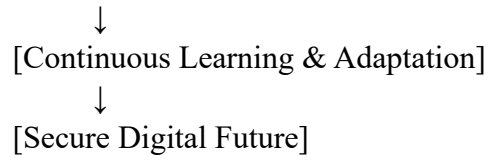


Figure 10. Relationship between evolving cyber threats, cybersecurity measures, continuous learning, and achieving a secure digital future

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